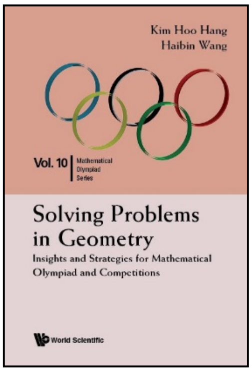
Olympiad Geometry – I

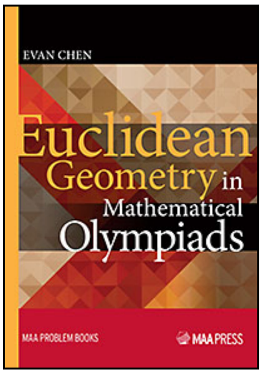
Some reading materials:

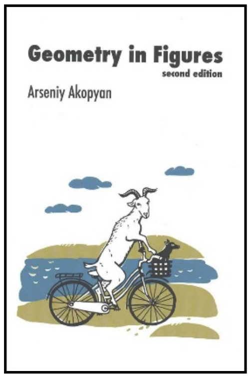
- Solving Problems in Geometry
  Insights and Strategies for Mathematical Olympiad and Competitions
  https://www.worldscientific.com/doi/pdf/10.1142/9789814583756_0001

- Euclidean Geometry in Mathematical Olympiads – Chapter 2 Circles
  https://www.maa.org/sites/default/files/pdf/ebooks/pdf/EGMO_chapter2.pdf

- Geometry in Figures
  https://www.mccme.ru/~akopyan/papers/EnGeoFigures.pdf







0. Notation and Basic Facts

a, b and c are the sides of ΔABC opposite to A, B, and C respectively.
$[ABC]$ = area of ΔABC

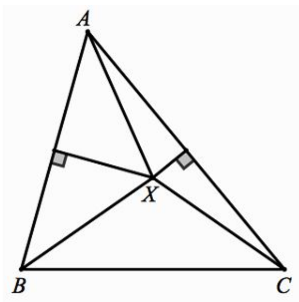
s = semi-perimeter $=\frac{1}{2}(a+b+c)$ r = inradius R = circumradius

Ravi Substitution: $x = s - a$, $y = s - b$, $z = s - c$
i.e. $a = y + z$, $b = x + z$, $c = x + y$.

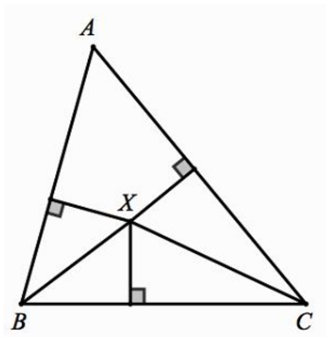
h_a, h_b, h_c – lengths of the altitudes drawn to the sides a, b, c respectively
m_a, m_b, m_c – lengths of the medians drawn to the sides a, b, c respectively
i_a, i_b, i_c – lengths of the bisectors of the angles A, B, C respectively

Centres of a triangle

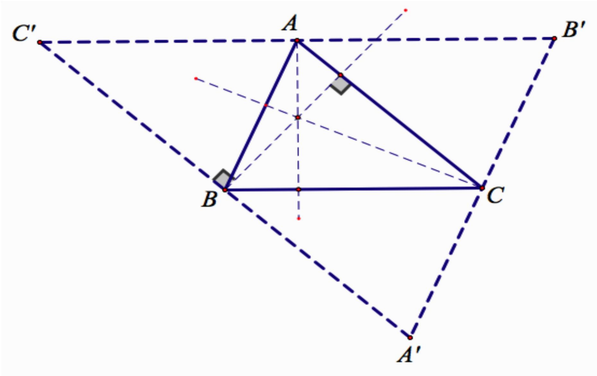
(a) **Circumcentre**
In a triangle, the perpendicular bisectors of the three sides meet at a point. This point is called the circumcentre of the triangle, usually denoted by O, and it is the centre of the circumcircle of the triangle.



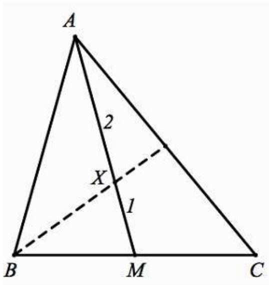
(b) **Incentre**
In a triangle, the internal angle bisectors meet at a point. This point is called the incentre of the triangle, usually denoted by I, and it is the centre of the incircle of the triangle.



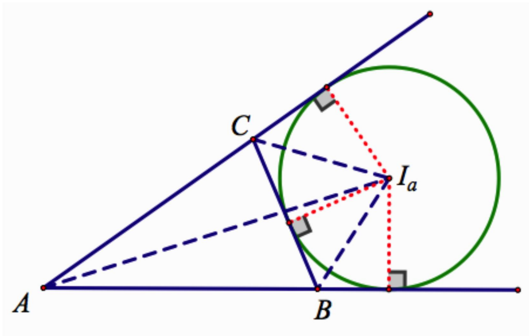
(c) **Orthocentre**
In a triangle, the altitudes meet at a single point. This point is called the orthocentre of the triangle and is usually denoted by H.



(d) **Centroid**
In a triangle, the medians meet at a point. This point is called the centroid of the triangle and is usually denoted by G.



(e) **Excentres**
In triangle ABC the A-angle bisector and the bisectors of external angle B and C meet at a point. This point is called the A-excentre of ΔABC, usually denoted by I_a and it is the centre of the A-excircle (circle tangent to the side BC and to the extended sidelines AB and AC). Similarly, we define points I_b and I_c.



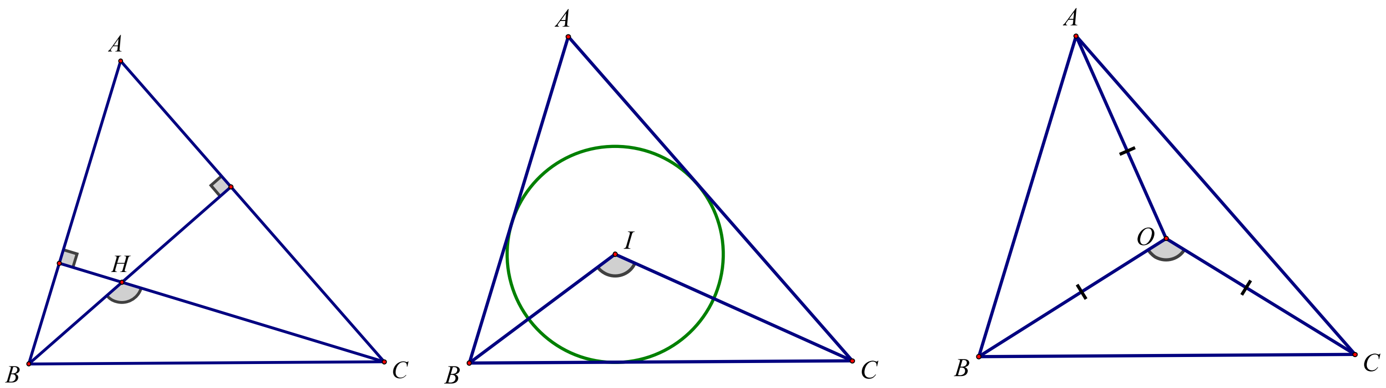
Angle at centres

Let ABC be a triangle with orthocentre H, incentre I and circumcentre O. Then:

(a) If triangle ABC is acute, then $\angle BHC = 180^\circ - \angle A$.

(b) $\angle BIC = 90^\circ + \dfrac{1}{2}\angle A$.

(c) If $\angle A$ is acute, then $\angle BOC = 2\angle A$.

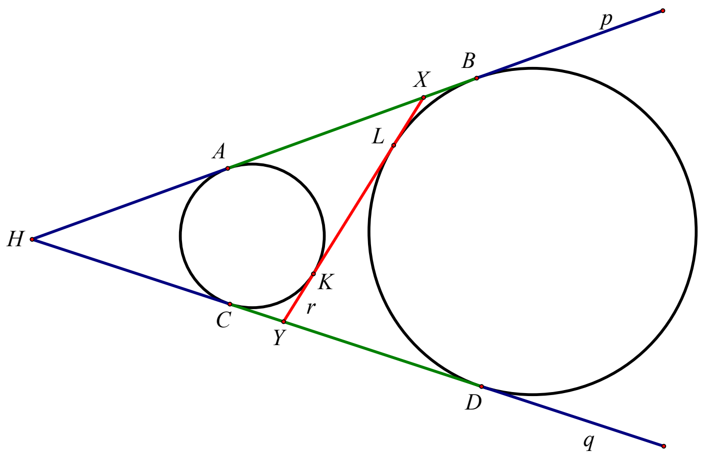


Lengths of tangents

Let p, q be common external tangents of circles ω_1 and ω_2. Denote by A, B the points of tangency of p with ω_1 and ω_2 respectively, and by C, D the points of tangency of q with ω_1 and ω_2 respectively. Then:

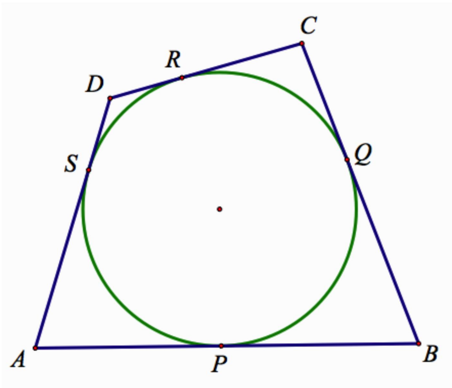
(a) $AB = CD$.

(b) If the two circles are nonintersecting and their common internal tangent r intersects p and q at points X, Y, respectively, we have $AB = CD = XY$.



Example 0.1 (Pilot's Theorem) [lengths of tangents]

Let $ABCD$ be a quadrilateral with an inscribed circle. Then $AB + CD = BC + DA$.



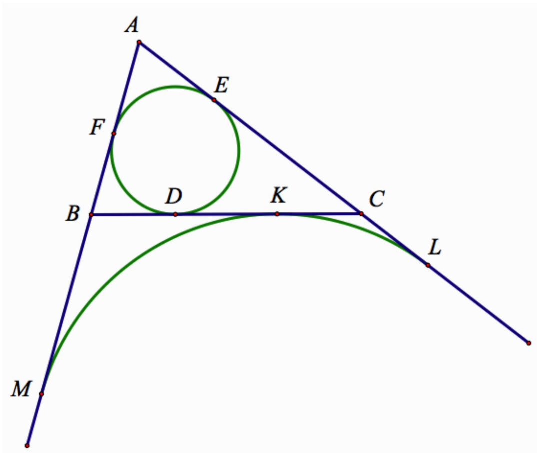
Segments of triangle related to tangency

Let ABC be a triangle with semiperimeter s. Denote by D, E, F the points of tangency of the incircle with sides BC, CA, AB respectively. Also let the A-excircle touch the lines BC, CA, AB at points K, L, M respectively. Then the following holds:

(a) $AE = AF = s - a$; $BD = BF = s - b$; $CD = CE = s - c$

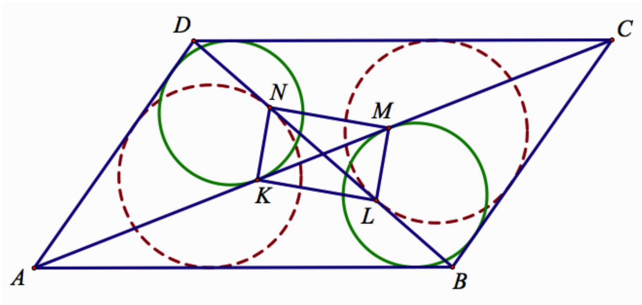
(b) $AL = AM = s$; hence $CL = CK = s - b$ and $BK = BM = s - c$

(c) Points K and D are symmetric with respect to the midpoint of BC.



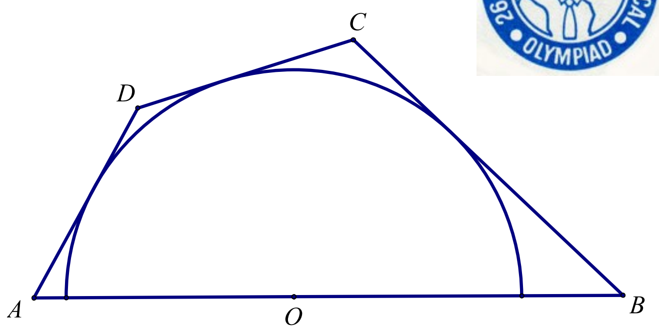
Example 0.2 [segments lengths related to tangency]

Let $ABCD$ be a parallelogram with $AB > BC$. Let K, M be the points of tangency of the incircles of triangles ACD and ABC with AC, respectively. Similarly, let L, N be the points of tangency of the incircles of triangles BCD and ABD with BD respectively. Prove that $KLMN$ is a rectangle.

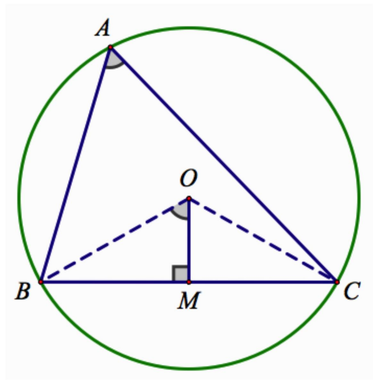


Example 0.3 (IMO 1985 P1)

A circle has centre on the side AB of the cyclic quadrilateral $ABCD$. The other three sides are tangent to the circle. Prove that $AD + BC = AB$.



1. Metric Relations



(Extended) Sine Formula: $\dfrac{a}{\sin A} = \dfrac{b}{\sin B} = \dfrac{c}{\sin C} = 2R$

Cosine Formula: $a^2 = b^2 + c^2 - 2bc\cos A$

Areas of triangle

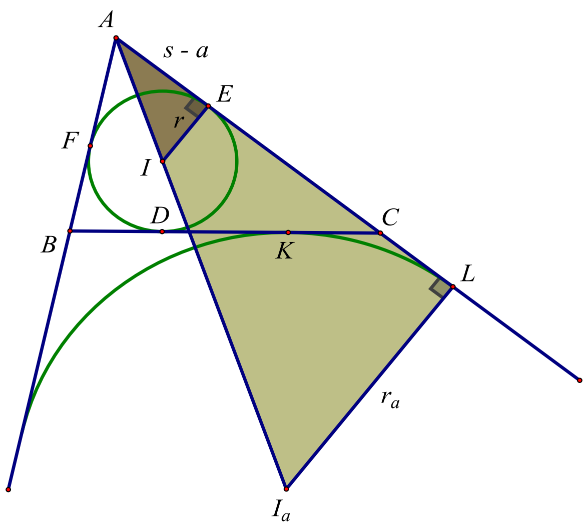
Let ABC be a triangle. Then we can calculate its area in the following ways:

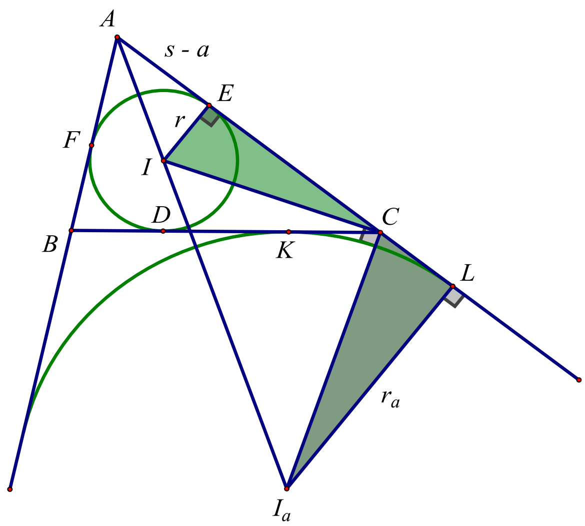
(a) $[ABC] = \dfrac{1}{2}ab\sin C = \dfrac{1}{2}bc\sin A = \dfrac{1}{2}ac\sin B = \dfrac{abc}{4R}$

(b) $[ABC] = \sqrt{s(s-a)(s-b)(s-c)} = \sqrt{sxyz}$ (Heron's Formula)

(c) $[ABC] = sr = r_a(s-a) = r_b(s-b) = r_c(s-c)$

(d) Let M be a point on the line BC and AM makes an angle θ (either acute or obtuse) with BC, then $[ABC] = \dfrac{1}{2}AM \times BC \times \sin\theta$.





***xyz* formula**

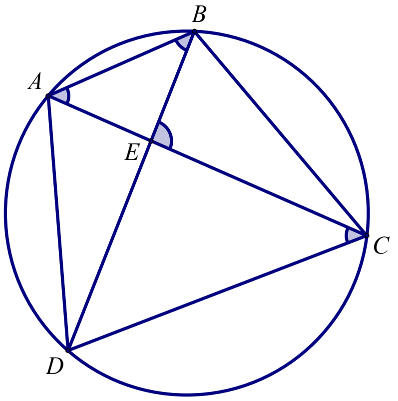
In triangle ABC, we can find the area $[ABC]$, inradius r, and circumradius R in terms of xyz as follows:

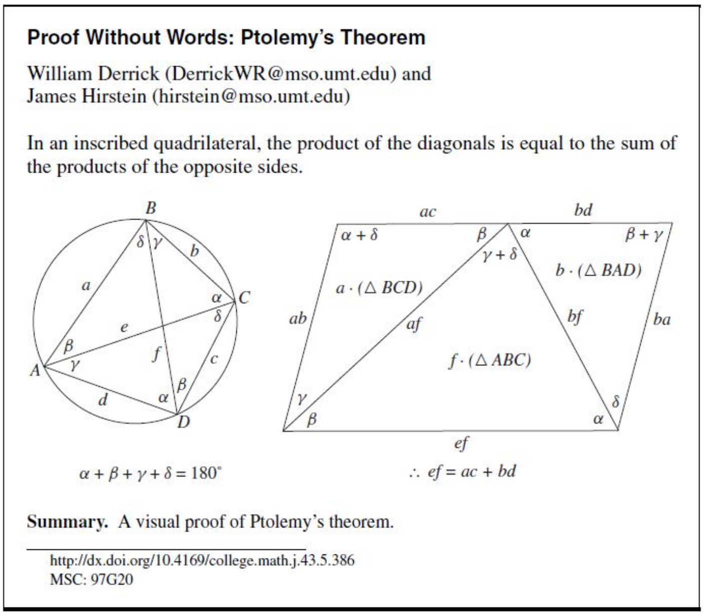
(a) $[ABC] = \sqrt{(x+y+z)xyz}$ (b) $r = \sqrt{\dfrac{xyz}{x+y+z}}$ (c) $R = \dfrac{(y+z)(z+x)(x+y)}{4\sqrt{xyz(x+y+z)}}$

Ptolemy's Theorem

If the convex quadrilateral *ABCD* is inscribed in a circle, then

$$AB \times CD + AD \times BC = AC \times BD.$$





Proof Without Words: Ptolemy's Theorem

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In an inscribed quadrilateral, the product of the diagonals is equal to the sum of the products of the opposite sides.

$\alpha + \beta + \gamma + \delta = 180^\circ$

$\therefore\ ef = ac + bd$

Summary. A visual proof of Ptolemy's theorem.

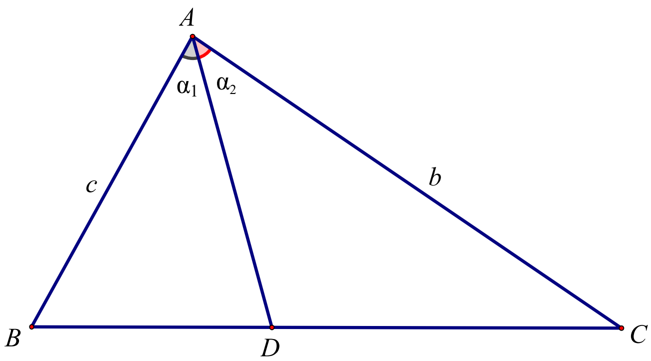
http://dx.doi.org/10.4169/college.math.j.43.5.386
MSC: 97G20

Proof:

Ratio Lemma

In triangle *ABC*, let $D \in BC$ and denote by α_1 and α_2 the angles *BAD* and *DAC* respectively. Then

$$\frac{BD}{DC} = \frac{AB \sin \alpha_1}{AC \sin \alpha_2}.$$

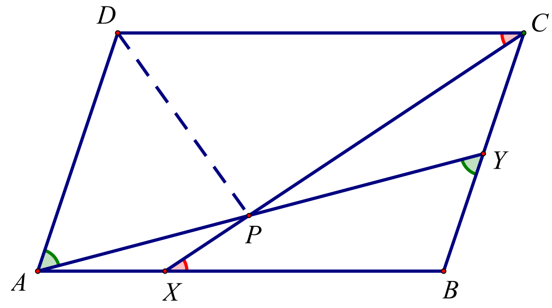


Angle Bisector Theorem

In triangle *ABC* let *AD*, $D \in BC$, be the internal angle bisector. Then

$$\frac{BD}{CD} = \frac{c}{b}, \qquad BD = \frac{ac}{b+c}, \qquad CD = \frac{ab}{b+c}$$

Example 1.1 (Germany 2003)
Let $ABCD$ be a parallelogram. Let X and Y be points on the sides AB and BC respectively, such that $AX = CY$. Prove that the intersection of lines AY and CX lies on the angle bisector of $\angle ADC$.

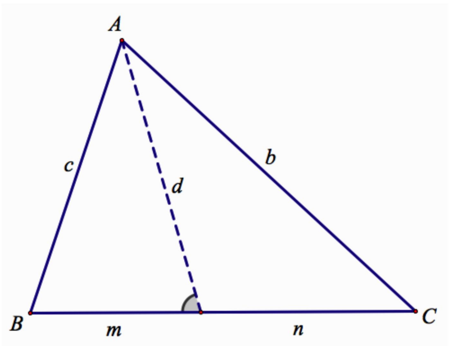


Example 1.2 (Criterion for orthogonality)
Let AC and BD be two lines in plane. Then $AC \perp BD$ if and only if $AB^2 + CD^2 = AD^2 + BC^2$.

Stewart's Theorem

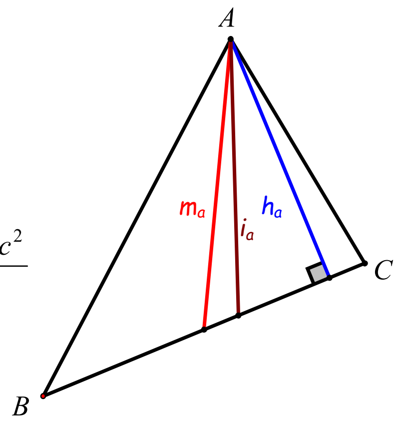
In triangle ABC let D lie on the side BC. Denote the distance BD, DC and AD by m, n and d respectively.

Then $d^2 + mn = b^2\left(\dfrac{m}{m+n}\right) + c^2\left(\dfrac{n}{m+n}\right)$



Proof:

Exercises:



1. Prove that $h_a = \dfrac{bc}{2R}$, $h_b = \dfrac{ac}{2R}$, $h_c = \dfrac{ab}{2R}$.

2. Prove that $m_a^2 = \dfrac{2(b^2+c^2)-a^2}{4}$, $m_b^2 = \dfrac{2(a^2+c^2)-b^2}{4}$, $m_c^2 = \dfrac{2(a^2+b^2)-c^2}{4}$

3. Prove that $i_a^2 = bc\dfrac{[(b+c)^2-a^2]}{(b+c)^2} = 4bc\dfrac{s(s-a)}{(b+c)^2}$ $i_b^2 = ca\dfrac{[(c+a)^2-b^2]}{(c+a)^2} = 4ca\dfrac{s(s-b)}{(c+a)^2}$
$i_c^2 = ab\dfrac{[(a+b)^2-c^2]}{(a+b)^2} = 4ab\dfrac{s(s-c)}{(a+b)^2}$.

4. **(IMO Preliminary Contest 2021 Problem 20)**
In ΔABC, $AB = 15$ and $AC = 13$. $DEFG$ is a square such that D is on AC, F is on AB and G is the mid-point of BC. If $AD = 11$ and $AF > FB$, find the area of $DEFG$.